Supplementary material for: A reproducible computational framework for decoupled wildfire ignition and propagation benchmarking in the Gran Chaco (2018–2025)

D2 target construction, calibration, evaluation protocol, extended tables and figures

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Code: github.com/sanchezmartin093-code/Incendios_PY (MIT License; code concept DOI 10.5281/zenodo.23675104; GitHub release v2.0.0).
Data: processed data and test predictions are available in the v2.0.0 GitHub release under CC BY 4.0 and indexed by the data concept DOI 10.5281/zenodo.23675278.

Supplementary contents

- **Note S1:** Data, D2 target construction, quality control and sampling.
- **Note S2:** Bounded monotonic Platt calibration and hyperparameters.
- **Note S3:** Evaluation protocol and extended analyses.
- **Tables S1–S16:** variable dictionary, hyperparameters, departmental results, computational profile, conformal coverage, population evaluation, paired significance, spatial blocks, lead time, sampling, full benchmark, ablation, M1 baseline, SHAP, cost-loss and data audit.
- **Figures S1–S3:** SHAP, ablation and cost-loss.

Supplementary Note S1. Data, D2 target construction, quality control and sampling

VIIRS active-fire detections from Suomi-NPP, NOAA-20 and NOAA-21 are acquired as 375-m cross-track swaths from NASA FIRMS (NASA FIRMS, 2026). At the swath edge the projected pixel can reach approximately 1.6 km along scan and 0.8 km along track, producing the geometric Bow-Tie effect: repeated detections of the same fire, dilution of fire radiative power (FRP) over a larger apparent area and incorrect assignment to the national 1-km^2 grid (Schroeder et al., 2014). The implemented kinematic filter is

$$\mathcal{D}_{\text{filtered}} = \{f \in \mathcal{D}_{\text{raw}} \mid \text{scan}(f) \leq 1.5 \text{ km} \ \wedge \ \text{confidence}(f) \neq \text{'low'}\}\,.$$

The observed maximum scan in the study window was 0.8 km, so the rule removed no records but remains active for orbital windows with larger view angles. Duplicate overpasses on the same civil date are consolidated by 1-km^2 cell, retaining `num_focos_dia`, `max_frp_dia` and `sum_frp_dia`.

D2 separates primary ignition from tactical propagation. Let F_t be the set of cells with a thermal detection on day t. D2 ignition at cell i and day t requires (i) $i \in F_t$, (ii) no cell in F_{t-1} within 25 km and (iii) no detection in cell i during the preceding 10 days. D2 propagation requires $i \in F_t$ and at least one cell in F_{t-1} within 25 km. The two target sets are disjoint. The 25-km threshold defines the target domain and is not a negative-sampling buffer; no 20-km buffer is applied to the test set. The 20-km buffer is used only in the spatial block holdout.

The case-control sample is drawn at 1:10 per task and region. Training uses 2018–2022, validation uses 2023 and the test is the frozen 2024–2025 period. Population denominators are kept separately and are used for prior correction. The same test rows are used for every model, FWI and INPE, so comparisons are paired. Sampling counts and class ratios are in Table S10. The variable dictionary is in Table S1; the full audit log is in Table S15.

Data quality checks collapse duplicate VIIRS overpasses, remove `low`-confidence detections, reconstruct daily precipitation from sub-daily increments and reject impossible values above 300 mm/day. ERA5-Land nodes (Muñoz-Sabater et al., 2021) are mapped to cells with a precomputed cKDTree. The processed data archive contains test targets, probabilities, benchmark layers, sampling plans and trained artifacts, so the result tables can be regenerated without the raw multi-terabyte archive.

Supplementary Note S2. Bounded monotonic Platt calibration and hyperparameters

Let $z_i \in \mathbb{R}$ be the margin produced by the LightGBM ensemble (Ke et al., 2017) for cell i, and let $y_i \in \{0, 1\}$ be the binary fire label. Standard Platt scaling (Platt, 1999) estimates $w, b \in \mathbb{R}$

by minimizing binary cross-entropy:

$$\mathcal{L}(w,b) = -\sum_{i=1}^{M} \left[y_i \log \sigma(wz_i + b) + (1 - y_i) \log(1 - \sigma(wz_i + b)) \right], \qquad \sigma(v) = \frac{1}{1 + e^{-v}}.$$

If and only if $w > 0$, the map $p(z) = \sigma(wz + b)$ is strictly increasing and preserves the ranking of the ensemble. Its derivative is

$$\frac{dp}{dz} = w\, \sigma(wz + b)\, [1 - \sigma(wz + b)],$$

and the sigmoid factor is strictly positive, so $\operatorname{sign}(dp/dz) = \operatorname{sign}(w)$. A positive slope therefore leaves ROC-AUC unchanged. Unconstrained logistic regression can converge to $w < 0$ under extreme imbalance or bimodal margins, inverting the ranking and turning a ROC-AUC of 0.77 into 0.23.

The implementation minimizes a bounded Brier loss with L-BFGS-B in `src/modelo.py`:

$$\min_{w,b} \frac{1}{M} \sum_{i=1}^{M} \left(y_i - \frac{1}{1 + \exp(-(wz_i + b))} \right)^2, \qquad 0.10 \le w \le 10.00, \quad -10.0 \le b \le 10.0.$$

The lower bound guarantees strict monotonicity and prevents gradient collapse in asymptotic regions. Calibration is fitted on 2023 and frozen for 2024–2025. The same cutoff logic is used for M1, M2, M3 and propagation; cutoffs are selected on validation with an F0.5 criterion and applied after prior correction. Hyperparameters are in Table S2. The ablation uses a common capacity (45/60 rounds and 23/31 leaves) to isolate the feature effect.

Supplementary Note S3. Evaluation protocol and extended analyses

The harness is fixed before the test is opened. It contains five components:

(1) A strictly temporal split: training 2018–2022, validation 2023 and test 2024–2025.

(2) A 50-km spatial block holdout with a 20-km buffer; priors are recomputed without held-out fires.

(3) A paired bootstrap grouped by date ($B = 1\,000$) for confidence intervals and differences against FWI (Van Wagner, 1987) and INPE (Setzer et al., 2019).

(4) A lead-time analysis using t-1 weather to quantify the nowcast-to-forecast gap.

(5) Population-level, conformal and cost-loss analyses.

Bootstrap resampling is by date, not by row, because daily fire activity is strongly clustered. All cells from a sampled date are retained; this preserves within-day dependence. The spatial holdout removes an entire 50-km block plus a 20-km buffer, recomputes historical priors without the held-out fires and evaluates both held-in and held-out cells. The primary metrics are ROC-AUC and PR-AUC; Brier score, expected calibration error, precision, recall, F0.5 and false positives are also reported. AUC intervals use 500 grouped bootstrap replicates; paired differences use 1 000. Because four main comparisons are made against FWI and INPE, the significance statements are a conservative screening result rather than multiplicity-adjusted inference. The bootstrap resolution is 0.001.

The full benchmark is in Table S11, paired significance in Table S7, the complete spatial block results in Table S8 and the lead-time results in Table S9. The controlled ablation is in Table S12 and Figure S2; SHAP is in Table S13 and Figure S1; population evaluation is in Table S6; conformal coverage is in Table S5; cost-loss is in Table S14 and Figure S3. The computational profile is in Table S4. All analyses use the frozen test rows archived with the code release.

The 1-km^2 grid is the modeling unit. Fuel fractions come from annual MapBiomas Collection 2 maps (2017–2023), joined to each row with the prior-year map; 2024 uses 2023 with lag 1 and 2025 reuses 2023 with lag 2 (MapBiomas Chaco, 2023). Historical priors use a causal three-year rolling window rather than the full 2018–2022 record. `rafaga_max_kmh` is a linear proxy and is not used as a feature. M3 contains 44 variables in the Chaco and 45 in the Oriental Region; M2 contains 35 and M1 contains 24.

Table S1. Variable dictionary and physical role of the predictors.

Variable	Description	Source	Role
temp_max_c	Daily maximum temperature at 2 m (°C)	ERA5-Land	Ignition and propagation
temp_min_c	Daily minimum temperature at 2 m (°C)	ERA5-Land	Ignition and propagation
rango_termico_diario	Intraday thermal range (Tmax − Tmin)	Derived	Ignition and propagation
vpd_max_kpa	Maximum vapor pressure deficit (kPa)	Derived	Gate, ignition and propagation
deficit_hidrico_acumulado	Integrated desiccation stress (VPD × (CDD + 1), kPa · d)	Derived	Ignition and propagation
precipitacion_mm	Daily rainfall (00–24 h, mm)	ERA5-Land	Ignition and propagation
precipitacion_48h_mm	Antecedent 48-h rainfall (mm)	Derived	Gate, ignition and propagation

Variable	Description	Source	Role
dias_secos_consecutivos	Consecutive days with rainfall <1.0 mm (CDD)	Derived	Ignition and propagation
viento_max_kmh	Maximum 10-m wind speed (km/h)	ERA5-Land	Ignition and propagation
rafaga_max_kmh	Linear wind proxy (1.3 × wind); not an ERA5-Land gust	Proxy	Not used
potencial_viento_cuadratico	Dynamic wind pressure ($(v/10)^2$, dimensionless)	Derived	Ignition and propagation
interaccion_viento_vpd	Advective–desiccant coupling (wind × VPD)	Derived	Ignition and propagation
swvl1_m3m3	Topsoil volumetric water content, 0–7 cm (m^3/m^3)	ERA5-Land	M1, M2, M3 and propagation
fdd_heladas_60d	Antecedent frost severity over 60 days (°C-days)	ERA5-Land; derived	M1, M2, M3 and propagation
deficit_hidro_60d	Antecedent water deficit over 60 days (mm)	ERA5-Land; derived	M1, M2, M3 and propagation
riesgo_estero_seco	Dry estero risk (wetland × 60-day deficit)	Derived	M1, M2, M3 and propagation
combustible_curado_helada	Frost-cured pasture and grassland fuel	Derived	M1, M2, M3 and propagation
ffmc	Fine Fuel Moisture Code	FWI derived	Ignition and propagation
isi	Initial Spread Index	FWI derived	Ignition and propagation
dia_semana	Civil weekday (0 = Monday … 6 = Sunday)	Calendar	Ignition and propagation
es_fin_de_semana	Weekend indicator (Saturday/Sunday)	Calendar	Ignition and propagation
mes	Month of year (1–12)	Calendar	Ignition and propagation
dia_del_anio	Ordinal day of year (1–366)	Calendar	Ignition and propagation
pct_pastura	Total pasture, annual agriculture and forest plantation (0–100 %)	Derived from MapBiomas C2 annual (Y-1)	INPE benchmark only; not a model feature

Variable	Description	Source	Role
pct_pastura_ganadera	Cultivated pasture fraction (0–100 %)	MapBiomas C2 annual (Y-1)	M1, M2, M3 and propagation
pct_agricultura_anual	Annual agriculture fraction (0–100 %)	MapBiomas C2 annual (Y-1)	M1, M2, M3 and propagation
pct_plantacion_forestal	Forest plantation fraction (0–100 %)	MapBiomas C2 annual (Y-1)	M1, M2, M3 and propagation
pct_sabana_pastizal	Savanna and natural grassland fraction (0–100 %)	MapBiomas C2 annual (Y-1)	Ignition and propagation
pct_humedal_palmar	Wetland and palm swamp fraction (0–100 %)	MapBiomas C2 annual (Y-1)	Ignition and propagation
pct_bosque_seco	Dry Chaco forest fraction (0–100 %)	MapBiomas C2 annual (Y-1)	Ignition and propagation (Chaco)
pct_bosque_humedo	Atlantic humid forest fraction (0–100 %)	MapBiomas C2 annual (Y-1)	Ignition and propagation (Oriental)
indice_ecotono	Forest–pasture edge disruption index	Derived	Ignition and propagation
borde_bosque_agro_pct	Forest–agriculture edge fraction (0–100 %)	Derived from MapBiomas C2 annual (Y-1)	M2, M3 and propagation
pct_desmonte_t1	Tree-cover loss in the prior year (0–100 %)	Hansen GFC-2025 v1.13 (Y-1)	M2, M3 and propagation
pct_desmonte_t2	Tree-cover loss two years before (0–100 %)	Hansen GFC-2025 v1.13 (Y-2)	M2, M3 and propagation
presion_desmonte_vpd	Deforestation × VPD pressure	Derived (Hansen Y-1 × VPD)	M2, M3 and propagation
dist_foco_hist_km	Minimum distance to training fires (km)	FIRMS, causal 3-year window	Ignition and propagation

Variable	Description	Source	Role
densidad_hist_25km	Cumulative historical fire density within 25 km	FIRMS, causal 3-year window	Ignition and propagation
dist_foco_t1_km	Minimum distance to an active t-1 fire (km)	Causal spatial kernel	Propagation only
focos_25km_t1	Active t-1 fires within 25 km	Causal spatial kernel	Propagation only
frp_25km_t1	Accumulated t-1 FRP within 25 km (MW)	Causal spatial kernel	Propagation only
dist_corredor_antropico_km	Distance to chronic-recurrence cells (3 events, 2018–2022)	Anthropogenic prior	M3 only
densidad_hist_5km	Kernel density of historical fires within 5 km	Anthropogenic prior	M3 only
densidad_hist_10km	Kernel density of historical fires within 10 km	Anthropogenic prior	M3 only
freq_hist_1km	Point recurrence in the 1-km^2 cell (2018–2022)	Anthropogenic prior	M3 Oriental and M2 fallback selector
temporada_quema_ganadera	Livestock burning season indicator (August–October)	Anthropogenic calendar	M3 only
doy_sin	Continuous seasonal sine harmonic	Derived	M3 only
doy_cos	Continuous seasonal cosine harmonic	Derived	M3 only
presion_quema_pastura	Pasture burning pressure (pasture × VPD × month weight)	Human–climate interaction	M3 only
focalizacion_antropica_local	Local-to-macro concentration ratio (5 km / 25 km)	Multiscale ratio	M3 only
interaccion_acceso_vpd	Accessibility–desiccation synergy (exp(−d_hist/5) × VPD)	Human–climate interaction	M3 only

Variable	Description	Source	Role
es_combustible	Combustible biomass indicator	MapBiomas C2 annual (Y-1)	Biophysical gate

The operational models were not selected through a fully registered search. The table reports the configuration of the boosters trained by `scripts/06_entrenar_modelos.py` and `scripts/16_ablacion_modelo_hibrido.py`.

Table S2. Operational and ablation hyperparameters.

Hyperparameter	M2 Chaco	Propagation Chaco	M2 Oriental	Propagation Oriental	M3 Chaco	M3 Oriental
objective	binary	binary	binary	binary	binary	binary
learning_rate	0.03	0.03	0.03	0.03	0.03	0.03
num_leaves	15	15	15	15	23	31
max_depth	5	5	5	5	5	5
min_child_samples	20	20	10	10	20	10
scale_pos_weight	1.0	1.0	2.0	2.0	1.0	2.0
feature_fraction	0.80	0.80	0.80	0.80	0.80	0.80
bagging_fraction	0.80	0.80	0.80	0.80	0.80	0.80
lambda_l1	0.20	0.20	0.20	0.20	0.20	0.20
lambda_l2	1.00	1.00	1.00	1.00	1.00	1.00
num_boost_round	20	20	40	40	45	60
early_stopping	None	None	None	None	None	None

The values were generated by `scripts/14_auditoria_suplementaria.py`.

Table S3. D2 test performance by department.

Department	Region	Sovereign cells	Test samples	Test fires	Ignition ROC-AUC	Propagation ROC-AUC	Tactical precision
SAN PEDRO	Oriental	20 854	22 859	2 763	0.7150	0.7590	60.7 %

Department	Region	Sovereign cells	Test samples	Test fires	Ignition ROC-AUC	Propagation ROC-AUC	Tactical precision
CONCEPCIÓN	Oriental	18 523	24 486	2 022	0.7050	0.8110	74.3 %
ITAPÚA	Oriental	15 868	21 861	1 355	0.7760	0.7800	50.4 %
CANINDEYÚ	Oriental	15 052	19 820	1 700	0.8510	0.8340	65.3 %
ALTO PARANÁ	Oriental	14 206	23 639	869	0.8430	0.8560	51.5 %
CAAGUAZÚ	Oriental	12 901	14 333	1 571	0.7260	0.7490	54.9 %
AMAMBAY	Oriental	12 555	15 122	1 517	0.7590	0.8360	76.1 %
ÑEEMBUCÚ	Oriental	11 672	15 506	1 555	0.6380	0.7190	54.8 %
CAAZAPÁ	Oriental	9 554	10 662	1 388	0.7130	0.7240	49.8 %
PARAGUARÍ	Oriental	8 593	9 631	1 139	0.6250	0.7030	57.0 %
MISIONES	Oriental	8 157	9 786	912	0.5970	0.7070	58.2 %
CORDILLERA	Oriental	4 791	4 947	561	0.6790	0.7310	55.2 %
GUAIRÁ	Oriental	3 866	4 455	469	0.6880	0.7160	46.8 %
CENTRAL	Oriental	2 406	2 751	347	0.7040	0.7780	68.0 %
ASUNCIÓN	Oriental	128	78	8	0.6800	0.7920	55.6 %
BOQUERÓN	Chaco	88 623	66 587	5 038	0.8080	0.9270	88.9 %
ALTO PARAGUAY	Chaco	78 613	60 983	4 075	0.8130	0.9210	86.9 %
PRESIDENTE HAYES	Chaco	73 644	49 046	6 943	0.7090	0.8550	80.0 %

ASUNCIÓN has only eight test fires; its department-level metrics are reported for completeness

and should not be interpreted as stable estimates.

The profile was measured on macOS ARM64; x86_64 was not measured. The point-in-polygon row is a reference for 50 000 points and is not used at inference.

Table S4. Stage-level profile for 2024-09-09.

Stage	Method	Cells	Time (s)	Peak RAM (MB)	Note
Integer cell_id lookup (O(1))	NumPy int64	400 006	0.00073	—	Median of 100 repetitions
Point-in-Polygon	GeoPandas sjoin	50 000	0.02610	—	50 000 points; non-linear extrapolation
Ingestion, priors and gate	Polars/cK-DTree	400 006	1.26800	—	Measured 2024-09-09
Dual inference	LightGBM C++	400 006	0.28500	—	Measured 2024-09-09
Aggregation and bulletin	Polars	400 006	0.01600	—	Measured 2024-09-09
Total pipeline	Integrated	400 006	1.56900	2 182.1	ARM64; x86_64 not measured

Table S5. Empirical conformal coverage. These values are not a formal guarantee under temporal shift.

Region	$\hat{q}_{90}$	Coverage 90 (%)	$\hat{q}_{95}$	Coverage 95 (%)
Chaco	0.3360	89.77 %	0.8457	95.04 %
Oriental	0.3743	90.18 %	0.8464	94.56 %

Population metrics apply prior correction from the case-control sample to the population denominator. Population precision is the relevant operational quantity, not case-control precision.

Table S6. Population-level evaluation with prevalence correction (retrospective 2024–2025 test).

Region	Task	Modelle	Pop. prevalence	Sample prevalence	Pop. cut-off	ROC-AUC	PR-AUC	Precision (%)	Recall (%)	F0.5	Brier	BSS	ECE	Pop. N	Pop. positives
Chaco	ignicion	M3_ignicion_hibrida	0.00015	0.5	0.9	0.900467	0.80591	0.0009	0.082	27.304	0.0010	0.00015	0.5	0.00030	0.00001	10497	1
Chaco	ignicion	M2_ignicion_base	0.00015	0.5	0.9	0.900428	0.80781	0.0007	0.090	24.589	0.0011	0.00015	0.5	0.00030	0.00001	10497	1
Chaco	propagacion	M2_propagacion_tactica	0.00156	0.5	0.9	0.922388	0.89088	0.1270	9.667	50.652	0.1153	0.00144	0.47666	0.00023	3374	8	5
Oriental	ignicion	M3_ignicion_hibrida	0.00041	0.4	0.9	0.901084	0.76270	0.0019	0.198	20.731	0.0025	0.00041	0.40005	0.00014	38873	1	6
Oriental	ignicion	M2_ignicion_base	0.00041	0.4	0.9	0.900955	0.77549	0.0015	0.166	22.106	0.0021	0.00041	0.40005	0.00004	38873	1	6
Oriental	propagacion	M2_propagacion_tactica	0.00210	0.2	0.9	0.917507	0.77963	0.0287	4.784	18.771	0.0562	0.00206	0.91135	0.00043	7235	15	9

Ign. = ignition; Prop. = propagation. M2 is the baseline ignition model in the ignition task and the tactical propagation model in the propagation task; M3 is the hybrid ignition model.

Table S7. Paired ROC-AUC differences against FWI and INPE, grouped by date (retrospective 2024–2025 test).

Region	Task	Comparison	ΔROC	p value	Bootstrap replicates
Oriental	ignicion	Incendios_PY (Ignición Híbrida M3) vs Canadian FWI completo	0.1970	0.001	1000
Oriental	ignicion	Incendios_PY (Ignición Híbrida M3) vs INPE Risco de Fogo	0.2544	0.001	1000
Chaco	ignicion	Incendios_PY (Ignición Híbrida M3) vs Canadian FWI completo	0.2848	0.001	1000
Chaco	ignicion	Incendios_PY (Ignición Híbrida M3) vs INPE Risco de Fogo	0.2633	0.001	1000

The comparison uses B = 1,000 grouped-by-date bootstrap replicates. Values are retrospective diagnostics, not a blind confirmation.

Table S8. Full spatial block results on 2023 development blocks.

Region	Model	Split	n	n positive	Prevalence	ROC-AUC	PR-AUC	Brier	BSS	ECE	Precision	Recall	F0.5	FP	FN
Chaco	M1_bio-rido	hold-out_espacial_2023	19304	1758	0.0910	0.6456	0.1668	0.0806	-0.0259	0.0065	0.1680	0.2685	0.1816	2337	1286
Chaco	M1_bio-rido	holdin_herencia_2023	78079	7095	0.0908	0.6280	0.1429	0.0812	-0.0176	0.0031	0.1611	0.2280	0.1712	8423	5477
Chaco	M2_base	hold-out_espacial_2023	19304	1758	0.0910	0.6773	0.1760	0.0843	-0.0183	0.0589	0.4828	0.0080	0.0374	15	1744
Chaco	M2_base	holdin_herencia_2023	78079	7095	0.0908	0.7832	0.2526	0.0748	0.0942	0.0049	0.3344	0.2148	0.3009	3033	5571
Chaco	M3_hi-brido	hold-out_espacial_2023	19304	1758	0.0910	0.6496	0.1630	0.0849	-0.0262	0.0606	0.4286	0.0085	0.0395	20	1743
Chaco	M3_hi-brido	holdin_herencia_2023	78079	7095	0.0908	0.7821	0.2511	0.0750	0.0926	0.0051	0.2761	0.3326	0.2858	6189	4735
Oriental	M1_bio-rido	hold-out_espacial_2023	25961	1998	0.0769	0.7176	0.1574	0.0683	0.0386	0.0123	0.1867	0.1957	0.1885	1703	1607
Oriental	M1_bio-rido	holdin_herencia_2023	65163	6286	0.0964	0.6899	0.1857	0.0835	0.0423	0.0009	0.2430	0.2140	0.2366	4190	4941
Oriental	M2_base	hold-out_espacial_2023	25961	1998	0.0769	0.7470	0.1820	0.0701	0.0137	0.0392	0.4286	0.0015	0.0074	4	1995

Region	Model	Split	n	n positive	Prevalence	ROC-AUC	PR-AUC	Brier	BSS	ECE	Precision	Recall	F0.5	FP	FN
Oriental	M2_base	held-in_reference_2023	65163	6286	0.0964	0.6736	0.2231	0.0814	-0.0659	0.0066	0.2628	0.2719	0.2646	4794	4577
Oriental	M3_hybrid	held-out_spatial_2023	25961	1998	0.0769	0.6277	0.1608	0.0716	-0.0075	0.0474	0.3750	0.0015	0.0074	5	1995
Oriental	M3_hybrid	held-in_reference_2023	65163	6286	0.0964	0.6747	0.2384	0.0806	-0.0757	0.0048	0.3087	0.2151	0.2840	3028	4934

M1 = biophysical only; M2 = macro-anchor baseline; M3 = multiscale socio-ecological hybrid. Holdout = 50-km spatial block; Held-in = reference cells not excluded from training. The confirmation holdout is not used here.

Table S9. Evaluation with t-1 weather (24-h anticipation) on 2023 validation.

Region	Model	n	n positive	ROC-AUC	PR-AUC	Brier	BSS	ECE	Cut-off F0.5	Precision	Recall	F0.5
Chaco	M1_biophysical_t1	97383	8853	0.6342	0.1467	0.0810	0.0201	0.0032	0.129195	0.1749	0.1799	0.1759
Chaco	M2_base_t1	97383	8853	0.7818	0.2444	0.0752	0.0902	0.0066	0.197258	0.2834	0.3063	0.2877
Chaco	M3_hybrid_t1	97383	8853	0.7831	0.2523	0.0749	0.0940	0.0039	0.219858	0.2935	0.2913	0.2931
Oriental	M1_biophysical_t1	91124	8284	0.6965	0.1765	0.0792	0.0412	0.0017	0.157237	0.2093	0.2703	0.2192
Oriental	M2_base_t1	91124	8284	0.7463	0.2167	0.0770	0.0677	0.0056	0.193620	0.2586	0.2572	0.2583
Oriental	M3_hybrid_t1	91124	8284	0.7613	0.2384	0.0759	0.0810	0.0038	0.213015	0.2852	0.2646	0.2808

M1 = biophysical only; M2 = base ignition; M3 = multiscale socio-ecological hybrid; t-1 = previous-day weather.

Table S10. D2 sampling counts and class ratios.

Partition	Task	Positives	Negatives	Ratio
Train	Ignition	91 271	912 710	1:10.00
Validation	Ignition	17 137	171 370	1:10.00
Test	Ignition	34 232	342 320	1:10.00
Train	Propagation	499 293	4 992 930	1:10.00
Validation	Propagation	75 706	757 060	1:10.00
Test	Propagation	130 674	1 306 740	1:10.00

The full benchmark includes M2, propagation, FWI, INPE and the climatological baseline. FWI and INPE are normalized indices, not calibrated probabilities.

Table S11. Full benchmark on the retrospective 2024–2025 test.

Region	Task	Model	ROC-AUC	PR-AUC	Brier	BSS	ECE	Precision	Recall	F0.5	FP
Chaco	ignicion	Incendios_PY (Ignición Híbrida M3)	0.8021	0.2842	0.0733	0.1131	0.0082	0.3282	0.3030	0.3228	9960
Chaco	ignicion	Incendios_PY (Ignición Base M2)	0.7991	0.2766	0.0737	0.1083	0.0104	0.3328	0.2725	0.3187	8773
Chaco	ignicion	Canadian FWI completo	0.5173	0.0957	—	—	—	0.0928	0.9608	0.1133	150807
Chaco	ignicion	INPE Risco de Fogo	0.5389	0.1031	—	—	—	0.1172	0.1257	0.1188	15207

Region	Task	Model	ROC-AUC	PR-AUC	Brier	BSS	ECE	Precision	Recall	F0.5	FP
Chaco	ignicion	Línea Base Climatológica	0.5000	0.0909	0.0826	0.0000	0.0000	0.0000	0.0000	0.0000	0
Chaco	propagacion	Incendios_PY (Propagación Táctica)	0.9039	0.6935	0.0442	0.4656	0.0086	0.8369	0.5240	0.7476	5353
Chaco	propagacion	Línea Base Climatológica	0.5000	0.0909	0.0826	0.0000	0.0000	0.0000	0.0000	0.0000	0
Oriental	ignicion	Incendios_PY (Ignición Híbrida M3)	0.7534	0.2340	0.0765	0.0748	0.0082	0.3013	0.1859	0.2680	7835
Oriental	ignicion	Incendios_PY (Ignición Base M2)	0.7406	0.2146	0.0774	0.0640	0.0070	0.2610	0.1958	0.2447	10075
Oriental	ignicion	Canadian FWI completo	0.5564	0.1061	—	—	—	0.1076	0.3899	0.1258	58766

Region	Task	Model	ROC-AUC	PR-AUC	Brier	BSS	ECE	Precision	Recall	F0.5	FP
Oriental	ignicion	INPE Risco de Fogo	0.4990	0.0908	—	—	—	0.0924	0.7947	0.1123	141802
Oriental	ignicion	Línea Base Climatológica	0.5000	0.0909	0.0826	0.0000	0.0000	0.0000	0.0000	0.0000	0
Oriental	propagacion	Incendios_PY (Propagación Táctica)	0.7859	0.3752	0.0691	0.1633	0.0092	0.6503	0.1984	0.4468	8351
Oriental	propagacion	Línea Base Climatológica	0.5000	0.0909	0.0826	0.0000	0.0000	0.0000	0.0000	0.0000	0

FWI = full Canadian Fire Weather Index; INPE = INPE Fire Risk; M2 = baseline ignition model in the ignition task and tactical propagation model in the propagation task; M3 = hybrid ignition model; Climatology = climatological baseline; Ign. = ignition; Prop. = propagation.

Table S12. Controlled ablation of M1, M2 and M3 on the retrospective 2024–2025 test.

Region	Architecture	Variables	ROC-AUC	PR-AUC	Brier	BSS	ECE	Precision (%)	Recall (%)	F0.5	FP
Chaco	M3: multi-scale socio-ecological hybrid	44	0.8021	0.2842	0.0733	0.1131	0.0082	32.81	30.31	0.3228	9963
Chaco	M2: macro-anchor ignition (v1)	35	0.7991	0.2766	0.0737	0.1083	0.0104	33.27	27.26	0.3187	8778
Chaco	M1: bio-physical only (no anthropic)	24	0.6421	0.1574	0.0806	0.0250	0.0045	17.78	18.22	0.1787	13525
Chaco	Benchmark: full Canadian FWI	6	0.5173	0.0957	—	—	—	9.28	96.08	0.1133	150807
Chaco	Benchmark: INPE Fire Risk	7	0.5389	0.1031	—	—	—	11.72	12.57	0.1188	15207

Region	Architecture	Variables	ROC-AUC	PR-AUC	Brier	BSS	ECE	Precision (%)	Recall (%)	F0.5	FP
Oriental	M3: multiscale socio-ecological hybrid	45	0.7534	0.2340	0.0765	0.0748	0.0082	30.13	18.59	0.2680	7835
Oriental	M2: macro-anchor ignition (v1)	35	0.7406	0.2146	0.0774	0.0640	0.0070	26.09	19.59	0.2447	10085
Oriental	M1: biophysical only (no anthropic)	24	0.6840	0.1660	0.0798	0.0345	0.0052	19.17	24.63	0.2006	18876
Oriental	Benchmark: full Canadian FWI	6	0.5564	0.1061	—	—	—	10.76	38.99	0.1258	58766
Oriental	Benchmark: INPE Fire Risk	7	0.4990	0.0908	—	—	—	9.24	79.47	0.1123	141802

M1 = biophysical only; M2 = macro-anchor baseline; M3 = multiscale socio-ecological hybrid; FWI = full Canadian Fire Weather Index; INPE = INPE Fire Risk. Feature counts are 24 (M1), 35 (M2), and 44/45 (M3 Chaco/Oriental).

Table S13. Mean absolute TreeSHAP values for the hybrid ignition model.

Region	Feature	Description	Mean abs. SHAP	Rank
Chaco	dist_foco_hist_km	Distance to historical ignition (km)	0.468108	1
Chaco	pct_bosque_seco	Dry Chaco forest fraction (%)	0.187779	2
Chaco	dist_corredor_antropico_km	Distance to active anthropogenic corridor (km)	0.080344	3
Chaco	focalizacion_antropica_Local	Local focalization index (5 km / 25 km)	0.077876	4
Chaco	pct_pastura	Cultivated pasture fraction (%)	0.034180	5
Chaco	doy_cos	Seasonal cosine harmonic (DOY)	0.032798	6
Chaco	potencial_viento_cuadratico	Dynamic wind pressure (v^2)	0.015865	7
Chaco	densidad_hist_5km	Local anthropogenic ignition density (5 km)	0.011482	8
Chaco	presion_quema_pastura	Pasture burning pressure (pasture × VPD × month)	0.010718	9
Chaco	indice_ecotono	Ecotone index	0.009495	10
Oriental	dist_foco_hist_km	Distance to historical ignition (km)	0.283984	1
Oriental	focalizacion_antropica_Local	Local focalization index (5 km / 25 km)	0.219708	2
Oriental	dist_corredor_antropico_km	Distance to active anthropogenic corridor (km)	0.118975	3
Oriental	pct_humedal_palmar	Palm/wetland fraction (%)	0.090180	4
Oriental	freq_hist_1km	Chronic recurrence in a 1-km^2 cell	0.064126	5
Oriental	pct_sabana_pastizal	Savanna/natural grassland fraction	0.060748	6
Oriental	pct_pastura	Cultivated pasture fraction (%)	0.034921	7
Oriental	ffmc	Fine fuel moisture code (FFMC)	0.024360	8
Oriental	dia_del_anio	Day of year (DOY)	0.020410	9
Oriental	indice_ecotono	Ecotone index	0.018477	10

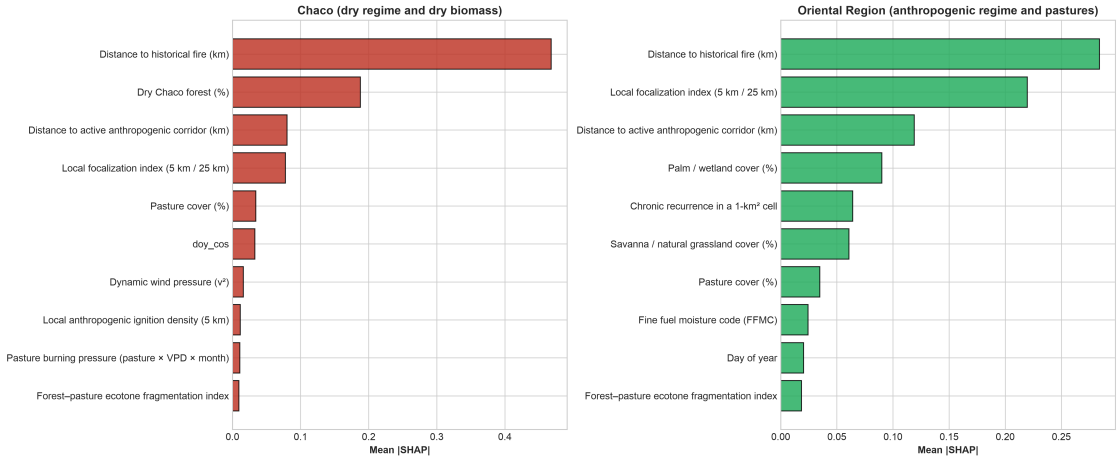


Figure 1: **Figure S1.** TreeSHAP comparison for the hybrid ignition model. The two regions use separate models and feature sets.

$\alpha = C/L$ is the ratio of false-alarm cost to avoided loss; $V(\alpha)$ is the normalized value. This is a simple decision model, not an estimate of agency budgets. TP, FP and FN are expressed as fractions of the population; rows with $V(\alpha) = 0$ are retained for completeness. M2 = baseline ignition model in the ignition task and tactical propagation model in the propagation task; M3 = hybrid ignition model.

Table S14. Cost-loss values by region, task, model and α.

Region	Task	Model	α	$V(\alpha)$	TP rate	FP rate	FN rate	Prevalence
Chaco	Ign.	M3	0.0001	0.353363	0.000136	0.478232	0.000017	0.000153
Chaco	Ign.	M3	0.0002	0.345244	0.000100	0.238185	0.000052	0.000153
Chaco	Ign.	M3	0.0005	0.053152	0.000032	0.047998	0.000121	0.000153
Chaco	Ign.	M3	0.0010	0.002592	0.000001	0.000603	0.000152	0.000153
Chaco	Ign.	M3	0.0020	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.0050	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.0100	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.0200	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.0300	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.0500	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.1000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.1500	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.2000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.3000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.4000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M3	0.5000	0.000000	0.000000	0.000000	0.000153	0.000153

Region	Task	Model	α	$V(\alpha)$	TP rate	FP rate	FN rate	Prevalence
Chaco	Ign.	M2	0.0001	0.355114	0.000136	0.474100	0.000017	0.000153
Chaco	Ign.	M2	0.0002	0.311610	0.000096	0.243186	0.000057	0.000153
Chaco	Ign.	M2	0.0005	0.013008	0.000024	0.044086	0.000129	0.000153
Chaco	Ign.	M2	0.0010	0.000366	0.000000	0.000011	0.000153	0.000153
Chaco	Ign.	M2	0.0020	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.0050	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.0100	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.0200	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.0300	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.0500	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.1000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.1500	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.2000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.3000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.4000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Ign.	M2	0.5000	0.000000	0.000000	0.000000	0.000153	0.000153
Chaco	Prop.	M2	0.0001	0.000000	0.001553	0.998447	0.000000	0.001553
Chaco	Prop.	M2	0.0002	0.000000	0.001553	0.998447	0.000000	0.001553
Chaco	Prop.	M2	0.0005	0.318867	0.001504	0.582993	0.000049	0.001553
Chaco	Prop.	M2	0.0010	0.524845	0.001247	0.168573	0.000306	0.001553
Chaco	Prop.	M2	0.0020	0.622603	0.001057	0.045164	0.000496	0.001553
Chaco	Prop.	M2	0.0050	0.540023	0.000933	0.018858	0.000620	0.001553
Chaco	Prop.	M2	0.0100	0.480229	0.000871	0.012425	0.000682	0.001553
Chaco	Prop.	M2	0.0200	0.405757	0.000795	0.008064	0.000758	0.001553
Chaco	Prop.	M2	0.0300	0.350540	0.000751	0.006695	0.000801	0.001553
Chaco	Prop.	M2	0.0500	0.263214	0.000640	0.004402	0.000912	0.001553
Chaco	Prop.	M2	0.1000	0.100916	0.000253	0.000871	0.001299	0.001553
Chaco	Prop.	M2	0.1500	0.024609	0.000079	0.000233	0.001473	0.001553
Chaco	Prop.	M2	0.2000	0.000494	0.000001	0.000003	0.001551	0.001553
Chaco	Prop.	M2	0.3000	0.000000	0.000000	0.000000	0.001553	0.001553
Chaco	Prop.	M2	0.4000	0.000000	0.000000	0.000000	0.001553	0.001553
Chaco	Prop.	M2	0.5000	0.000000	0.000000	0.000000	0.001553	0.001553
Oriental	Ign.	M3	0.0001	0.108055	0.000403	0.781304	0.000011	0.000414
Oriental	Ign.	M3	0.0002	0.223223	0.000374	0.576666	0.000040	0.000414
Oriental	Ign.	M3	0.0005	0.309159	0.000244	0.231920	0.000170	0.000414
Oriental	Ign.	M3	0.0010	0.126219	0.000108	0.055401	0.000306	0.000414
Oriental	Ign.	M3	0.0020	0.035039	0.000033	0.009109	0.000381	0.000414
Oriental	Ign.	M3	0.0050	0.002940	0.000003	0.000311	0.000411	0.000414
Oriental	Ign.	M3	0.0100	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.0200	0.000000	0.000000	0.000000	0.000414	0.000414

Region	Task	Model	α	$V(\alpha)$	TP rate	FP rate	FN rate	Prevalence
Oriental	Ign.	M3	0.0300	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.0500	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.1000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.1500	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.2000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.3000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.4000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M3	0.5000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.0001	0.098205	0.000404	0.795252	0.000011	0.000414
Oriental	Ign.	M2	0.0002	0.213427	0.000376	0.597393	0.000038	0.000414
Oriental	Ign.	M2	0.0005	0.299350	0.000254	0.260173	0.000160	0.000414
Oriental	Ign.	M2	0.0010	0.096056	0.000089	0.049283	0.000325	0.000414
Oriental	Ign.	M2	0.0020	0.009330	0.000009	0.002552	0.000405	0.000414
Oriental	Ign.	M2	0.0050	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.0100	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.0200	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.0300	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.0500	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.1000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.1500	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.2000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.3000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.4000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Ign.	M2	0.5000	0.000000	0.000000	0.000000	0.000414	0.000414
Oriental	Prop.	M2	0.0001	0.000000	0.002102	0.997898	0.000000	0.002102
Oriental	Prop.	M2	0.0002	0.000000	0.002102	0.997898	0.000000	0.002102
Oriental	Prop.	M2	0.0005	0.112058	0.002057	0.795132	0.000045	0.002102
Oriental	Prop.	M2	0.0010	0.220486	0.001856	0.532198	0.000246	0.002102
Oriental	Prop.	M2	0.0020	0.388701	0.001285	0.202092	0.000817	0.002102
Oriental	Prop.	M2	0.0050	0.242135	0.000702	0.038460	0.001400	0.002102
Oriental	Prop.	M2	0.0100	0.168701	0.000506	0.014958	0.001596	0.002102
Oriental	Prop.	M2	0.0200	0.112571	0.000378	0.006948	0.001724	0.002102
Oriental	Prop.	M2	0.0300	0.081204	0.000286	0.003735	0.001816	0.002102
Oriental	Prop.	M2	0.0500	0.040467	0.000170	0.001612	0.001932	0.002102
Oriental	Prop.	M2	0.1000	0.004147	0.000012	0.000029	0.002090	0.002102
Oriental	Prop.	M2	0.1500	0.000000	0.000000	0.000000	0.002102	0.002102
Oriental	Prop.	M2	0.2000	0.000000	0.000000	0.000000	0.002102	0.002102
Oriental	Prop.	M2	0.3000	0.000000	0.000000	0.000000	0.002102	0.002102
Oriental	Prop.	M2	0.4000	0.000000	0.000000	0.000000	0.002102	0.002102
Oriental	Prop.	M2	0.5000	0.000000	0.000000	0.000000	0.002102	0.002102

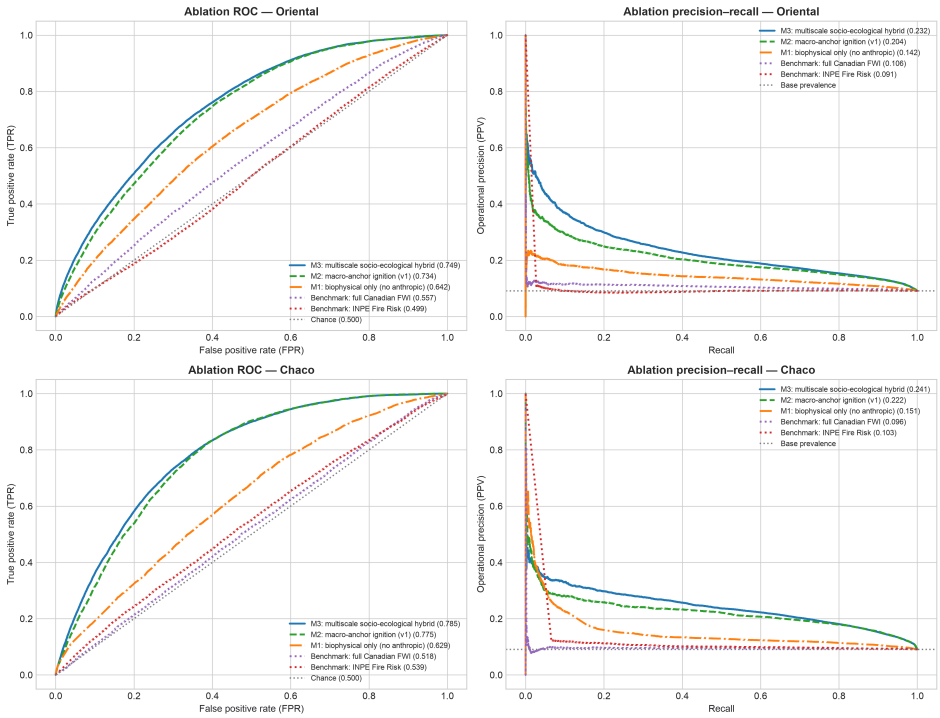


Figure 2: **Figure S2.** Controlled ablation under the D2 test. The figure complements Table S12.

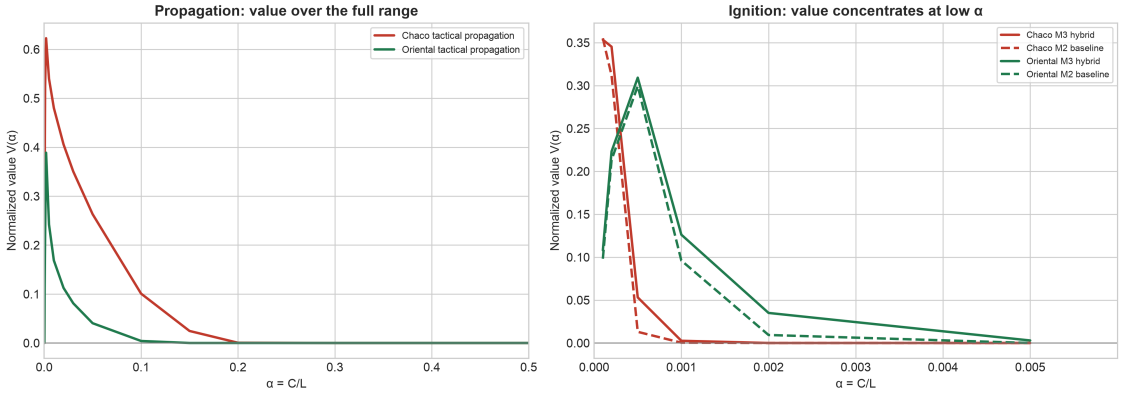


Figure 3: **Figure S3.** Cost-loss curves. The propagation panel uses the full α range; the ignition panel is truncated at $\alpha = 0.006$ because the decision value is zero or negligible at larger values. The figure complements Table S14.

Table S15. Data audit log generated by the repository checks.

Section	Check	Value	Status	Note
Grid	Rows	400 006	OK	
Grid	Unique cell IDs	400 006	OK	
Grid	Chaco cells	240 880	OK	
Grid	Oriental cells	159 126	OK	
Climate	Rows	1 870 080	OK	
Climate	Nodes	640	OK	
Climate	First date	2018-01-01	OK	
Climate	Last date	2025-12-31	OK	May exceed the climate window.
Climate	Precipitation >300 mm	0	OK	Plausible but unlikely; review de-accumulation.
Climate	Maximum precipitation (mm)	237.383	OK	External Open-Meteo correction is still pending.
Climate	Nonzero ET0	1 870 080	OK	Hargreaves-Samani from CDS Tmax/Tmin; not FAO-56.
Climate	Gust proxy 1.3×	4e-06	WARN	Gust is a 1.3× wind proxy.
FIRMS	Positive cell-days	892 533	OK	
FIRMS	First date	2018-01-01	OK	
FIRMS	Last date	2026-01-04	WARN	May exceed the climate window.
Raw FIRMS	Detections	3 685 836	OK	
Raw FIRMS	Maximum scan (km)	0.8	WARN	The Bow-Tie filter retains records when maximum scan 1.5 km.
Raw FIRMS	Detections with type != 0	5319	WARN	Type was not filtered; review industrial/offshore sources.
Unified dataset	Rows	915 561 114	OK	
Unified dataset	Fires	867 430	OK	

Section	Check	Value	Status	Note
Unified dataset	Prevalence	0.000947	OK	
Unified dataset	Cells	397 448	OK	
Partitions	Train ignition rows	1 003 981	OK	
Partitions	Train ignition prevalence	0.090909	OK	
Priors	Train Chaco positives at zero historical distance (%)	48.17	OK	100 % would indicate leakage from the event itself.
Priors	Train Oriental positives at zero historical distance (%)	62.79	OK	100 % would indicate leakage from the event itself.
Partitions	Validation ignition rows	188 507	OK	
Partitions	Validation ignition prevalence	0.090909	OK	
Priors	Validation Chaco positives at zero historical distance (%)	59.75	OK	100 % would indicate leakage from the event itself.
Priors	Validation Oriental positives at zero historical distance (%)	78.85	OK	100 % would indicate leakage from the event itself.
Partitions	Test ignition rows	376 552	OK	
Partitions	Test ignition prevalence	0.090909	OK	
Priors	Test Chaco positives at zero historical distance (%)	58.78	OK	100 % would indicate leakage from the event itself.
Priors	Test Oriental positives at zero historical distance (%)	71.92	OK	100 % would indicate leakage from the event itself.
Spatial leakage	Test cells present in training (%)	90.65	WARN	Temporal test; spatial holdout requires blocks or regions.
Models	M3 Chaco booster	present	OK	
Models	M3 Oriental booster	present	OK	
Models	Propagation Chaco booster	present	OK	

Section	Check	Value	Status	Note
Models	Propagation Oriental booster	present	OK	
Manifest	Pending hashes	0	OK	
Manifest	Rows	69	OK	

Table S16. M1 biophysical baseline on 2023 validation.

Region	Evaluation	ROC-AUC	PR-AUC	F0.5
Chaco	Temporal 2023, all cells	0.6354	0.1480	0.1762
Oriental	Temporal 2023, all cells	0.7041	0.1818	0.2301
Chaco	Cells without history (`freq_hist_1km = 0`)	0.6650	0.0841	—
Oriental	Cells without history (`freq_hist_1km = 0`)	0.7018	0.0740	—

M1 is a diagnostic biophysical baseline and was not used operationally. Spatial block results are in Table S8, lead-time results in Table S9, and the retrospective 2024–2025 ablation in Table S12.

Data availability

Processed data, test predictions, trained models and derived tables are available in the v2.0.0 GitHub release under CC BY 4.0 and indexed by the data concept DOI 10.5281/zenodo.23675278. Raw NASA FIRMS, ERA5-Land/CDS, MapBiomas, Hansen GFC and geoBoundaries data are obtained from their original providers under their own terms and are documented with SHA-256 hashes in `fuentes/MANIFIESTO_DATOS_v3.csv`. The archived test targets and probabilities are sufficient to regenerate every table and figure in this supplement without downloading the raw multi-terabyte archive.

Computer code availability

Name: `Incendios_PY`
Developer and contact: Martín Sánchez Insfrán, `sanchez.martin093@gmail.com`
Year first available: 2026
Hardware: x86_64 or ARM64, at least 8 GB RAM
Operating system: macOS 13+, Linux Ubuntu 22.04+, Windows 11 with WSL2
Language: Python 3.12+; Polars, DuckDB and NumPy for vectorized operations; LightGBM C++ backend
Dependencies: pinned in `requirements-lock.txt`
License: MIT
Repository: https://github.com/sanchezmartin093-code/Incendios_PY
Code concept DOI: 10.5281/zenodo.23675104
Tests: 76 tests pass with `python -m pytest -q tests`. The repository includes a README in English, installation and usage instructions, a user guide, and the scripts required to reproduce the main results.

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